BCR-6605 – edits to make

#	File	Where
1	WCTreeCellModifier	`modify()` – threshold branch
2	HealthAlarmsPreferenceStore	`innerSave()` threshold loop
3	HealthAlarmsPreferenceStore	`saveDeviations()` + new helper
4	HealthAlarmsComposite	`performApply()`
5	WCModelDisplayProperty	enum list
6	WCThresholdModel	new field
7	WCThresholdBaseline	new field
8	WCTreeCellModifier	new `case POINTS:`
9	HealthAlarmsPreferenceStore	`getThresholdModels()` ×2, helper, DTO
10	IndicatorDeviation + Dto + DB	new column
11	HealthCommonBusiness	`createDeviations`, `toDtos`
12	AbstractFilteringRuleAlgorithm	field, setter, `getFilteredValues`

Deviation columns (no popup, no configuration): Level, Type, Minimum, Maximum, Points, Filtering Law, TE.
Configuration columns (popup + configuration, as today): Value, Learning Time, LT Min Deviation, LT Mean, LT Dev Coef, Successive points, Max time w/o acquisition.

1 – WCTreeCellModifier

Method: `modify()`, branch `else if (data instanceof WCThresholdModel thresholdModel)`

1a. FIND

```
            } else if (data instanceof WCThresholdModel thresholdModel) {
               switch (WCModelDisplayProperty.fromKey(property)) {
```

REPLACE

```
            } else if (data instanceof WCThresholdModel thresholdModel) {
               boolean configurationField = false;
               switch (WCModelDisplayProperty.fromKey(property)) {
```

1b. Add `configurationField = true;` as the first line of these 7 cases only:

```
CHECKBOX   VALUE   LEARN_TIME   MIN_DEV   MEAN_MULT_FACTOR   DEV_COEF   NB_POINTS
```

Example:

```
                  case VALUE:
                     configurationField = true;
                     if (value != null && !value.toString().trim().isEmpty()
```

Do not add it to LEVEL, TYPE, MIN_VALUE, MAX_VALUE, WORKCARD_REF.

1c. FIND (3rd `markAsDirty`, end of this branch)

```
               MgsPreferenceStoreManager.markAsDirty((Aircraft) treeViewer.getData("aircraft"));
               treeViewer.refresh();
            }
         } catch (ParseException e) {
```

REPLACE

```
               if (configurationField) {
                  MgsPreferenceStoreManager.markAsDirty((Aircraft) treeViewer.getData("aircraft"));
               }
               treeViewer.refresh();
            }
         } catch (ParseException e) {
```

2 – HealthAlarmsPreferenceStore.innerSave()

Where: the `for (WCThresholdModel wcThresholdModel : indicatorThresholdModels)` loop

REPLACE the whole loop

```
               for (WCThresholdModel wcThresholdModel : indicatorThresholdModels) {
                  if (wcThresholdModel.isEnabled()) {
                     // BCR-6605 deviation fields from the baseline, configuration fields as edited
                     final WCThresholdBaseline baseline = wcThresholdModel.getBaseline() != null
                           ? wcThresholdModel.getBaseline()
                           : new WCThresholdBaseline(wcThresholdModel, wcThresholdModel.getFilteringLawName());
                     final ThresholdConfigurationDto thresholdConfigurationDto;
                     if (baseline.getType() == ThresholdType.FIXED) {
                        thresholdConfigurationDto = new ThresholdConfigurationDto(
                              wcThresholdModel.getSeverity(), baseline.getLevel(), baseline.getType(),
                              wcThresholdModel.getDefaultValue(),
                              wcThresholdModel.getNbSuccessivePoints(),
                              wcThresholdModel.getMinimumEngineOilTemperature());
                     } else {
```

```
                        thresholdConfigurationDto = new ThresholdConfigurationDto(
                              wcThresholdModel.getSeverity(), baseline.getLevel(), baseline.getType(),
                              wcThresholdModel.getLearningTime(),
                              baseline.getMinValue(), baseline.getMaxValue(),
                              wcThresholdModel.getDefaultValue(), wcThresholdModel.getMinimumDeviation(),
                              wcThresholdModel.getMeanFactor(), wcThresholdModel.getDeviationCoefficient(),
                              wcThresholdModel.getNbSuccessivePoints(),
                              wcThresholdModel.getMinimumEngineOilTemperature());
                     }
                     thresholdConfigurationDto.setWorkcard(baseline.getWorkCardReference());
                     indicatorConfigurationDto.getThresholdConfigurationDtos().add(thresholdConfigurationDto);
                  }
               }
```

3 – HealthAlarmsPreferenceStore.saveDeviations()

3a. Add this method to the class

```
   private boolean hasDeviationFieldChanged(final WCThresholdModel model) {
      final WCThresholdBaseline baseline = model.getBaseline();
      if (baseline == null) {
         return false;
      }
      return !Objects.equals(baseline.getLevel(), model.getLevel())
            || !Objects.equals(baseline.getType(), model.getType())
            || !Objects.equals(baseline.getMinValue(), model.getMinValue())
            || !Objects.equals(baseline.getMaxValue(), model.getMaxValue())
            || !Objects.equals(baseline.getFilteringLawName(), model.getFilteringLawName())
            || !Objects.equals(baseline.getMaxPointsPerHour(), model.getMaxPointsPerHour())
            || (model.getTeReference() != null && !model.getTeReference().trim().isEmpty());
   }
```

3b. In `saveDeviations()`, FIND

```
            if (!model.isDeviated() || model.getDeviationId() != null) {
               continue;
            }
```

REPLACE

```
            if (!model.isDeviated() || model.getDeviationId() != null) {
               continue;
            }
            if (!hasDeviationFieldChanged(model)) {
               continue;
            }
```

3c. In `toDeviationDto()`, delete the lines that set these on the DTO:

```
learningTime   minimumDeviation   meanFactor   deviationCoefficient   nbSuccessivePoints
```

4 – HealthAlarmsComposite.performApply()

4a. FIND

```
      if (MgsPreferenceStoreManager.wasLastSaveConfirmed()) {
         healthAlarmsPreferenceStore.saveDeviations();
      }
```

REPLACE

```
      healthAlarmsPreferenceStore.saveDeviations();
      List<String> deviated = healthAlarmsPreferenceStore.getLastSavedDeviationIndicatorNames();
      if (!deviated.isEmpty()) {
         MessageDialog.openInformation(getShell(),
               Messages.CORE.getString("deviation.created.title"),
               Messages.CORE.getString("deviation.created.message") + String.join(", ", deviated));
      }
```

4b. Message bundle

```
deviation.created.title=Deviation created
deviation.created.message=Deviation created on indicator
```

4c. In `saveDeviations()`, collect the names and expose them

```
   private final List<String> lastSavedDeviationIndicatorNames = new ArrayList<>();

   public List<String> getLastSavedDeviationIndicatorNames() {
      return lastSavedDeviationIndicatorNames;
   }
```

Clear it at the start of `saveDeviations()`, and add `indicatorDisplayNames.get(model.getIndicator().getId())` to it for each DTO built.

5 – WCModelDisplayProperty

FIND

```
FILTERING_LAW("FilteringLaw", 100), TE("TeReference", 120);
```

REPLACE

```
FILTERING_LAW("FilteringLaw", 100), TE("TeReference", 120), POINTS("MaxPointsPerHour", 50);
```

Bundle: `MaxPointsPerHour=Points`. If the tree lists its columns explicitly rather than iterating `values()`, add POINTS there and give it the same cell editor as NB_POINTS.

6 – WCThresholdModel

```
   private Integer maxPointsPerHour;

   public Integer getMaxPointsPerHour() {
      return maxPointsPerHour;
   }

   public void setMaxPointsPerHour(Integer maxPointsPerHour) {
      this.maxPointsPerHour = maxPointsPerHour;
   }
```

7 – WCThresholdBaseline

Field, beside `filteringLawName`

```
   private final Integer maxPointsPerHour;
```

Last line of the constructor

```
      this.maxPointsPerHour = model.getMaxPointsPerHour();
```

Getter

```
   public Integer getMaxPointsPerHour() {
      return maxPointsPerHour;
   }
```

8 – WCTreeCellModifier, new case

Insert before `case NB_POINTS:`

```
                case POINTS:
                   if (value != null && !value.toString().trim().isEmpty()) {
                      Number val = format.parse(replaceSeparatorValue(value).toString());
                      setPointsOnBothSeverities(thresholdModel, val.intValue());
                   } else {
                      setPointsOnBothSeverities(thresholdModel, null);
                   }
                   break;
```

Add to the class

```
   // BCR-6605 maxPointsPerHour is per indicator, shown on the RED and AMBER rows
   private void setPointsOnBothSeverities(final WCThresholdModel edited, final Integer points) {
      edited.setMaxPointsPerHour(points);
      if (treeViewer.getInput() instanceof WCMechanicalComponentModel root) {
         applyPoints(root, edited.getIndicator(), points);
      }
   }

   private void applyPoints(final WCMechanicalComponentModel model, final Indicator indicator,
         final Integer points) {
      for (WCMechanicalComponentModel sub : model.getSubHealthWCModel()) {
         applyPoints(sub, indicator, points);
      }
      for (WCHealthMechanicalComponentModel hmc : model.getHealthComponentModels()) {
         for (WCThresholdModel threshold : hmc.getThresholdModels()) {
            if (indicator.equals(threshold.getIndicator())) {
               threshold.setMaxPointsPerHour(points);
            }
         }
      }
   }
```

9 – HealthAlarmsPreferenceStore, Points

9a. In `getThresholdModels()`, FIND

```
         String icId = indicatorConfiguration != null ? indicatorConfiguration.getId() : null;
         for (WCThresholdModel thresholdModel : indicatorThresholdsMap.values()) {
            thresholdModel.setFilteringLawName(law);
            thresholdModel.setIndicatorConfigurationId(icId);
            thresholdModel.setBaseline(new WCThresholdBaseline(thresholdModel, law));
```

REPLACE

```
         String icId = indicatorConfiguration != null ? indicatorConfiguration.getId() : null;
         Integer points = indicatorConfiguration != null ? indicatorConfiguration.getMaxPointsPerHour() : null;
         for (WCThresholdModel thresholdModel : indicatorThresholdsMap.values()) {
            thresholdModel.setFilteringLawName(law);
            thresholdModel.setIndicatorConfigurationId(icId);
            thresholdModel.setMaxPointsPerHour(points);
            thresholdModel.setBaseline(new WCThresholdBaseline(thresholdModel, law));
```

9b. Same method, after the `getNbSuccessivePoints()` block

```
               if (deviation.getMaxPointsPerHour() != null) {
                  thresholdModel.setMaxPointsPerHour(deviation.getMaxPointsPerHour());
               }
```

9c. In `toDeviationDto()`

```
      if (!Objects.equals(baseline.getMaxPointsPerHour(), model.getMaxPointsPerHour())) {
         dto.setMaxPointsPerHour(model.getMaxPointsPerHour());
      }
```

10 – IndicatorDeviation, IndicatorDeviationDto, DB

Same field in both classes (no `@Column` on the DTO):

```
   @Column(name = "MAX_POINTS_PER_HOUR")
   private Integer maxPointsPerHour;

   public Integer getMaxPointsPerHour() {
      return maxPointsPerHour;
   }

   public void setMaxPointsPerHour(Integer maxPointsPerHour) {
      this.maxPointsPerHour = maxPointsPerHour;
   }
```

```
ALTER TABLE public.indicator_deviation ADD COLUMN max_points_per_hour INTEGER;
```

11 – HealthCommonBusiness

`createDeviations`

```
         deviation.setMaxPointsPerHour(dto.getMaxPointsPerHour());
```

`toDtos`

```
         dto.setMaxPointsPerHour(d.getMaxPointsPerHour());
```

Also in `toDtos`: move the `getFilteringLaw().getContextualParameters()` loop inside the `if (d.getFilteringLaw() != null)` guard. It is outside it today.

12 – AbstractFilteringRuleAlgorithm

12a. Field + setter, after the constructor

```
   private Map<String, IndicatorDeviation> deviations = Collections.emptyMap();

   public void setDeviations(final Map<String, IndicatorDeviation> applicableDeviations) {
      this.deviations = applicableDeviations != null ? applicableDeviations : Collections.emptyMap();
   }
```

12b. In `getFilteredValues`, FIND

```
            if (conf.getMaxPointsPerHour() == null) {
               valuesToSave.addAll(healthIndicatorValueAdapterValues);
            } else {
               healthIndicatorValueAdapterValues.sort(INDICATOR_VALUE_COMPARATOR);
               int maxPointsPerHour = getMaxPointsAtKey(key, conf.getMaxPointsPerHour());
```

REPLACE

```
            Integer configuredPoints = conf.getMaxPointsPerHour();
            final Integer deviatedPoints = getDeviatedPoints(conf);
            if (deviatedPoints != null) {
               configuredPoints = deviatedPoints;
            }
            if (configuredPoints == null) {
               valuesToSave.addAll(healthIndicatorValueAdapterValues);
            } else {
               healthIndicatorValueAdapterValues.sort(INDICATOR_VALUE_COMPARATOR);
               int maxPointsPerHour = getMaxPointsAtKey(key, configuredPoints);
```

12c. Add to the class

```
   private Integer getDeviatedPoints(final IndicatorConfiguration conf) {
      if (deviations.isEmpty() || conf.getIndicator() == null) {
         return null;
      }
      for (Severity severity : Severity.values()) {
         IndicatorDeviation deviation = deviations.get(conf.getIndicator().getId() + "|" + severity.name());
         if (deviation != null && deviation.getMaxPointsPerHour() != null) {
            return deviation.getMaxPointsPerHour();
         }
      }
      return null;
   }
```

Imports: `java.util.Collections`, `com.eurocopter.mgs.datamodel.alarm.Severity`, `com.eurocopter.mgs.datamodel.health.monitoring.IndicatorDeviation`.

12d. Ctrl+H on `new HelionixFilteringPerFileAlgorithm` and `new MarmsFilteringPerFlightHoursAlgorithm`. After the object is created, add

```
      filteringAlgorithm.setDeviations(healthHistoryProvider.getDeviations());
```

If no provider is in scope there, skip this line.

Already specified elsewhere, still to do

☐	File	Edit
	CreateIndicatorChartModelsJob	import `IndicatorDeviationDto`
	CreateIndicatorChartModelsJob	`getApplicableDeviation()` after line 334
	CreateIndicatorChartModelsJob	`retrieveThresholdValues`, lines 1079–1120
	RedAmberAlarmAnalyzer	remove the work card override
	persistence.xml	`ddl-generation` = `none`

Test

#	Action	Expect
1	Edit Level, add TE, Apply	"Deviation created" popup only. Deviation row, no configuration row
2	Edit Value only, Apply	Applicable flight popup only. Configuration row, no deviation row
3	Edit both, Apply	Both popups, both rows. Configuration keeps the old Level
4	Edit Points on the RED row	AMBER row follows
5	`SELECT te_reference, max_points_per_hour FROM public.indicator_deviation ORDER BY start_date DESC LIMIT 2;`	Value stored
6	A/C configuration screen, Apply	Unchanged